Sabancı Universitesi **Syllabus Application**

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CS 303
Logic and Digital System Design

Faculty Faculty of Engineering and Natural Sciences

Semester Spring 2025-2026

Course CS 303 - Logic and Digital System Design

Time/Place

Time	Week Day	Place	Date
08:40-09:30	Mon	SBS-G071	Feb 16-May 22, 2026
12:40-14:30	Thu	SBS-G071	Feb 16-May 22, 2026

Level of course Undergraduate

Course Credits SU Credit:4, ECTS:7, Basic:1, Engineering:6

Prerequisites

Corequisites CS 303L

Course Type Lecture

Program Requirements >

Instructor(s) Information

Ayhan Bozkurt

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Course Information

Catalog Course Description

Number systems and conversion, Boolean algebra, Boolean function minimization techniques, combinational logic circuit design, state elements (flip- flops), sequential circuits, design and implementation of state machines, Mealy and Moore circuits, higher level digital system design using logic building blocks such multiplexers/decoders, adders, memory and programmable gate arrays, hardware description languages.

Course Learning Outcomes:

1.	Explain the reasons for using different formats to represent numerical data and how negative integers are stored in sign-magnitude and two?s-complement representation.
2.	Convert numerical data from one format or base to another.
3.	Describe the internal representation of nonnumeric data.
4.	Demonstrate an understanding of the basic building blocks such as logic gates, flip-flops, counters, registers, and programmable logic devices
5.	Demonstrate the ability to minimize logic expressions, and express Boolean functions in different forms and an understanding of the physical considerations of logic elements such as gate delays.
6.	Use mathematical expressions to describe the functions of simple combinational and sequential circuits.
7.	Design combinational and sequential circuits using the fundamental building blocks given the verbal description of the circuits.
8.	Construct a finite state diagram to capture state transition in a sequential circuit.
9.	Demonstrate an understanding of digital systems expressed in register transfer level.

Course Objective

To develop the engineering skills for designing digital systems.
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Course Materials

Resources:

Textbook:
• M. Morris Mano and Michael D. Ciletti. Digital Design, 5e/6e, Pearson.

Technology Requirements:

Computer Usage: Logic Circuit Simulators / FPGA Development Tools
• Digital : https://github.com/hneemann/Digital
• The Tang Nano 9K : https://wiki.sipeed.com/hardware/en/tang/Tang-Nano-9K/Nano-9K.html

Assessment

The final grade for this course will be based on:

	Percentage (%)	Number of assessment methods
Final	30%	
Midterm	40%	2
Individual Project	16%	4
Group Project	10%	1
Homework	4%	4

Exam Schedule:

Time	Event	Details
Apr 11, 2026 15:30	Midterm: CS303 Midterm I	
May 3, 2026 11:00	Midterm: CS303 Midterm II	

Policies

Course Policies:

• Homework and lab assignments are announced one week before the deadline.
• Submission deadlines are never postponed.
• Laboratory sessions might be attended in groups of two. No group shuffling.

- No makeups for laboratory sessions.
- Exam makeups to be held at the end of final exam week (scheduled by SR).
- A minimum grade of 20/100 from the final exam to pass CS303.

AI Policies

Guidelines on the "Use of Generative AI" for students

SU Academic Integrity Statement:

https://www.sabanciuniv.edu/en/academic-integrity-statement

Health Report Requirements

Student Medical Reports Instruction Letter

Weekly Content

Week	Topic
Week 1	Introduction to Digital Systems
Week 2	Number Systems & Arithmetic
Week 3	Boolean Algebra & Logic Operations
Week 4	Gate-Level Minimization
Week 5	Gate-Level Minimization
Week 6	Analysis & Design of Combinational Logic Circuits
Week 7	Analysis & Design of Combinational Logic Circuits
Week 8	Analysis & Design of Combinational Logic Circuits
Week 9	Analysis & Design of Synchronous Sequential Logic Circuits
Week 10	Analysis & Design of Synchronous Sequential Logic Circuits
Week 11	Registers & Counters
Week 12	Registers & Counters
Week 13	Design with Algorithmic State Machines (ASM)
Week 14	Memory & Design with Programmable Logic